Land Cover Analysis

Materials:

Name	Time Period	File Type	Source Link
Bordner Survey	~1928-38	Vector land use data	Link
Aerial Photographs	1961	Orthomosaic	Link
NLCD	Yearly 1985-2023	Classified land use raster	Link
NAIP	2022	Raster	Link

Bordner Survey

Forest Cover

Select by Attributes

Where COV1 is equal to: A1, A2, A3, A3u, A3uu, A4, B, B1, B2, B3, C1, C1b, C2, C3, D, D1, D1b, D1u, D1uu, D1uuu, D2, D3, D3b, D3u, D4, D4b, E1, KC, LO, MA, NF, RC

Agriculture

Select by Attributes

Where COV1 is equal to: C, P, PP

Grant Basin Orthomosaic

Tile layer package downloaded from IU3D, extracted package to ArcGIS Pro as a raster

Forest Cover

Object-Based Classification (Source)

1. Segmentation
2. Radiometric Correction
3. Random training segments
    a. Elevation
    b. Slope
    c. Transformed Aspect (1 + cos(45 - A) * pi/180)
    d. AET
    e. TWI
    f. Brightness

    g. Band 1
4. randomForest analysis (R)
5. Classification

Agriculture/Contour Cropping

? Filter

High-pass filter

? Classification

Object-Based Classification

? Fuzzy Objects

Potential Sources:

https://doi.org/10.1016/S0957-4158(00)00003-9

https://doi.org/10.1016/j.compag.2021.106281

https://doi.org/10.1016/j.jag.2013.05.002

DOI: 10.3141/2291-10

NLCD Data

Annual USGS NLCD

Extract by Mask

Apply Symbology

NAIP Aerial Imagery

Grant Basin

USGS EarthExplorer– 8/6/2022 & 8/30/2022

Mosaic to New Raster

Extract by Mask

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Methods

Land Cover

The study area was defined as the Grant and Kickapoo hydrologic basins within the Driftless region of Wisconsin. All data was clipped to the provided shapefiles of both basins using ArcGIS Pro 3.4.0. These basins span Grant, Crawford, Vernon, Richland, and Monroe counties. The Driftless region is defined by its unique lack of ice cover during the last glacial period, resulting in a lack of glacial till deposits (Carson et al. 2019).

To analyze local effects of land cover on flooding and flood peaks, land cover data were compiled and analyzed at ten instances between 1928 and 2024.[1] The data for this analysis was predominantly sourced from the United States Geological Survey's National Land Cover Database (USGS, NLCD), with 30-m resolution datasets published annually since 1985. NLCD rasters were selected with a frequency of five years (1985, 1990, 1995, etc.) and analyzed according to their composition of land cover types within the study area. The concentration of forest and agricultural cover were of particular interest due to their hydrologic impact. For forest cover, the NLCD classes 41, 42, 43, and 90 were selected.[2] Classes 81 (Pasture/Hay) and 82 (Cultivated Crops) were both selected as agricultural cover classes, yet they were kept distinct. Another source of land cover data was found in the Bordner Survey, an inventory of land use and land cover in Wisconsin performed in the field between the 1920s and 1940s (Wisconsin Land Economic Inventory). The digitized versions of the Survey's maps were filtered by the

[1] Snapshots from other points within this period may be included as addenda following accuracy assessment, including 1961.

[2] Deciduous Forest, Evergreen Forest, Mixed Forest, and Woody Wetlands, respectively.

established classes belonging to forest and agricultural cover, resulting in a selection of polygons that were then transformed into rasters[3].

The land area of the three land cover categories–forest, cropland, and hay/pasture– was calculated using the Zonal Geometry tool in ArcGIS Pro, then recorded and analysed using Microsoft Excel (V. 2601).This involved finding the total area of each class of interest and comparing that sum against the total area of the basins. With the NLCD data, for example, the area of classes 41, 41, 43, and 90 were added together to find the total area of forest cover. This process was repeated across all ten data snapshots for both study area basins. Then, in Excel, the percentage of total land area that each land cover category represents was calculated for each year, leading to a time series of land cover ratio data. The percent change between each study year was also calculated in Excel.

A series of difference analyses were performed to produce visualizations displaying how land cover has changed locally across the study period. Using Reclassify, rasters were made binary, where 1 indicated the target land cover category (either forest or agriculture) and 0 indicated all other land cover classes. With the Raster Calculator tool, the differences in land cover between 1928, 1985, and 2024 were calculated. The binary approach allowed for a distinct identification of loss and gain for both land cover categories. Another approach also began using Reclassify, but this time, both forest and agriculture cover were included as 1 and 2, respectively, with 0 still indicating all other land classes. After calculating the differences between these trinary rasters, the results were transformed into point classes, then Kriging was used to create a color-ramped surface raster indicating particular regions of afforestation and deforestation between years.

Orthomosaic Classification– GIS

Another source of land cover data for this study came in the form of orthomosaic imagery, or combined aerial imagery. The orthomosaics focused on in this study are from a collection of aerial photographs sourced from the U.S Department of Agriculture (USDA), dated to ca. 1961 and mosaicked by IU3D (State Cartographer's Office). Traditionally, supervised object-based classification methods can be applied to satellite or aerial imagery when the model

[3] For forest cover, classes A1, A2, A3, A3u, A3uu, A4, B, B1, B2, B3, C1, C1b, C2, C3, D, D1, D1b, D1u, D1uu, D1uuu, D2, D3, D3b, D3u, D4, D4b, E1, KC, LO, MA, NF, and RC were selected. For cultivated crops, class C was selected, and for hay and pasture, classes P and PP were selected.

is trained on ground truth data taken from the image in question. However, due to these images being taken in black-and-white, these traditional models struggle to correctly classify land cover categories. A set of 300 random points within the study area were manually assigned ground truth class values, outlined in Figure 1.

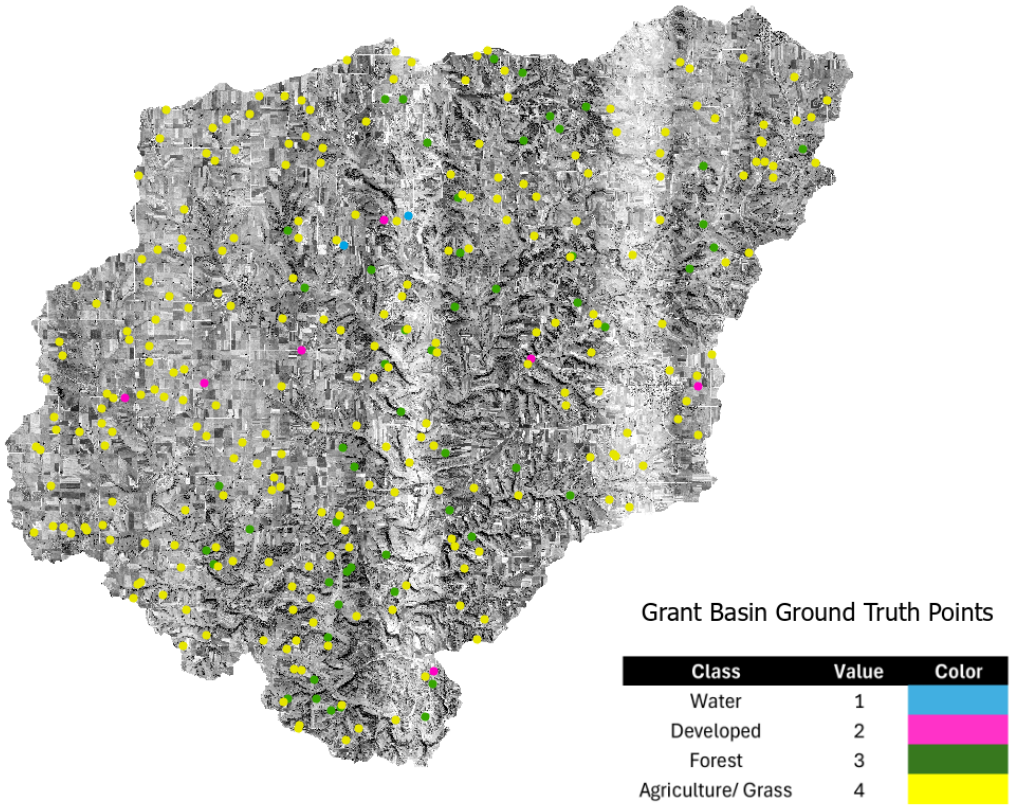

Figure 1. A map showing the 1961 Grant Basin orthomosaic image overlain by 300 ground truth points, identified by color.

Using these points, the highest overall accuracy achieved with the supervised classification modeling tools native to ArcGIS Pro was 84.67%, and the highest forest cover identification accuracy was 79.07%. In order to produce more accurate results, a pixel-based approach was used, requiring the creation of supplemental data.

Classification of the 1961 orthomosaic was accomplished using the randomForest package in R, which relies on a series of variables as predictors of class (Lydersen and Collins 2019). See Table 1 for a complete list of variables used. The variable Band 1 was simply an extraction of values from the original orthomosaic, without alteration. Brightness as a variable was calculated through methods of radiometric correction (Zhang et al. 2019). The average date of the images as well as the length of the shadows from a building of known height in roughly

the center of the mosaic was used to backcalculate the position of the sun when the images were taken.[4] By combining this knowledge with a Hillshade raster, a radiometrically averaged image was produced from which brightness values were extracted to the 300 ground truth points using the Extract Values to Points tool. Notably, this method of radiometric correction is very imprecise, as the orthoimages used to create the mosaic were taken across a week, and likely at different times of day. If the radiometric correction process were to occur as the mosaic were being constructed, then accuracy would be improved. Actual Evapotranspiration was found using the Monthly Actual Evapotranspiration (SSEBop) dataset from Senay et al. (2020). Unfortunately, these data were not available in 1961, so 2003 was chosen as the earliest available year.

The remainder of the variables were calculated from the local Digital Elevation Model from the Wisconsin State Cartographer's Office (State Cartographer's Office). Elevation was extracted from this raster as a variable, Z. Slope and aspect were calculated using their respective tools in ArcGIS Pro, with aspect transformed into units of radians (Lydersen and Collins 2018)[5]. A topographic wetness index (TWI) is a measurement of relative soil moisture based on topographic features (Beven and Kirkby 1979). To calculate this, a series of functions were performed, outlined in Figure 2.

Table 1. The title and data source for each of the seven variables used in the randomForest analysis and subsequent orthomosaic classification.

Variable	Source
Band 1	Orthomosaic
Brightness	Orthomosaic
Elevation (Z)	DEM
Slope	DEM
Transformed Aspect	DEM
Topographic Wetness Index (TWI)	DEM
Actual Evapotranspiration (AET)	SSEBop

[4] Solar geometry values were calculated using suncalc.org and the NOAA Solar Position Calculator.
[5] I think.

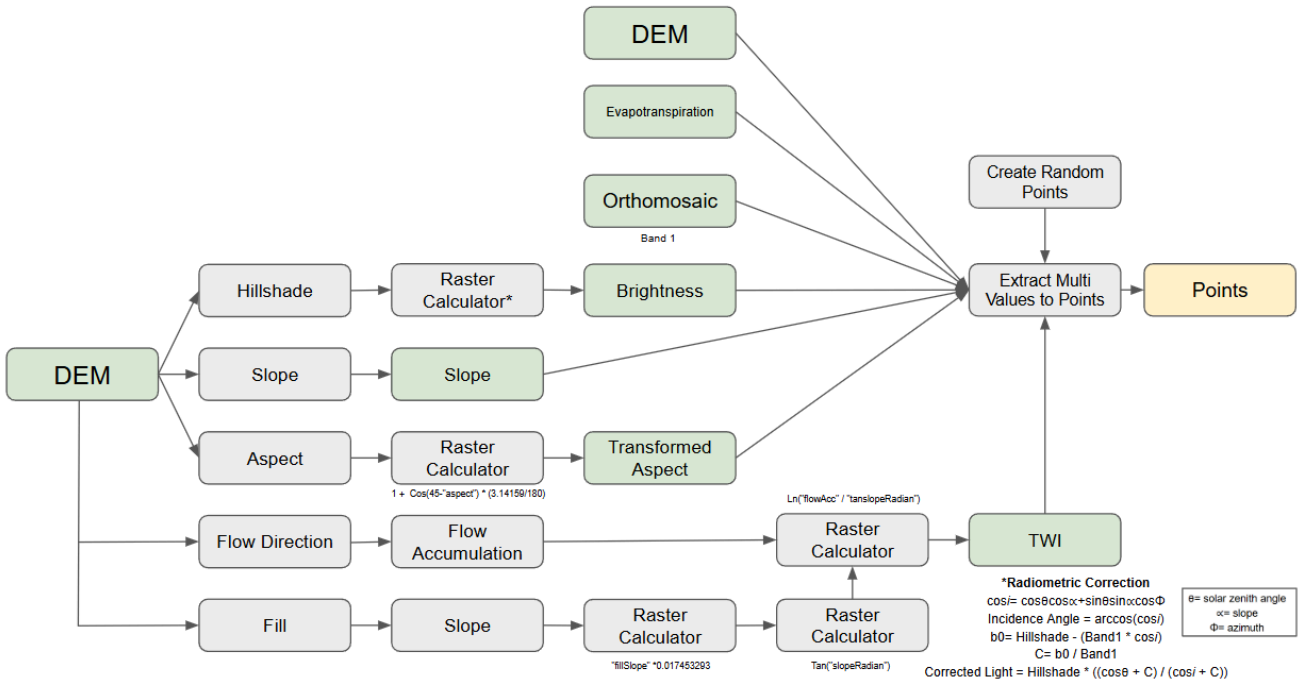

Figure 2. A flowchart describing the methods for calculating and extracting the values of each variable used for class prediction in the randomForest model. All steps were performed in ArcGIS Pro.[6]

These seven variables became the predictors for the randomForest analysis. In RStudio (2026.01.0), a table containing the values of each variable at each ground truth point was used as an input table. In order to create a raster containing the same variables as the input table, the Composite Bands tool in ArcGIS Pro was used to combine the six calculated variable rasters with the original orthomosaic image. The resultant raster was input into RStudio, where the classification scheme defined by randomForest prediction was applied.

Orthomosaic Classification– RStudio

Both the supplemental data creation and randomForest steps of the orthomosaic classification process have been compiled into a single R program.[7] Using this code, any black and white orthomosaic can be classified given the user provides the required input data. However, if the original orthomosaic or the final composite image are relatively large, it may be necessary for the

[6] Python code made available at https://github.com/annepett-commits/bw_orthomosaic_classification.
[7] Made available at https://github.com/annepett-commits/bw_orthomosaic_classification.

process to be handed off to a supercomputing server, if available. IU Bloomington's research supercomputer Quartz was employed for this project.

The final R script operates as a replica of the ArcGIS Pro methodology model seen in Figure 2. If the user is more comfortable working in ArcGIS Pro, the R script can also be used solely for the randomForest analysis step of the classification process if the orthomosaic, composite image, and ground truth point input table are first created using Esri or other GIS software. The R script begins by generating each of the variables found in Table 1 using the sf, terra, tidyterra, and whitebox packages. Any user of this R script must have on hand several files, listed in Table 2, that should be input into their designated spaces in the code.

Table 2. A list of all the data required for users to input into the black and white orthomosaic classification R script.

File	Data Type	Necessary For	Suggested Source
Orthomosaic	Raster	Band1, Brightness	
DEM	Raster	Slope, Aspect, TWI, Z	USGS EarthExplorer
AET	Raster	AET	SSEBop
Study Area Border	Shapefile		
Solar Zenith Angle & Azimuth	Numeric	Brightness	NOAA Solar Position Calculator

Available literature has suggested other variables may be of use in classifying black and white orthomosaics using randomTrees analysis, such as raster texture (Lydersen and Collins 2018). Further use of this methodology may explore this and other variables as potential inputs. Figure 3 demonstrates the relative importance or weight of each variable in this study's iteration of the classification process. Band 1 was selected as the most impactful variable in determining class, followed closely by slope. The remaining five variables fall within a similar range, with AET listed as the least impactful. Further iterations of this classification process may reveal which variables might be added or removed to increase accuracy.

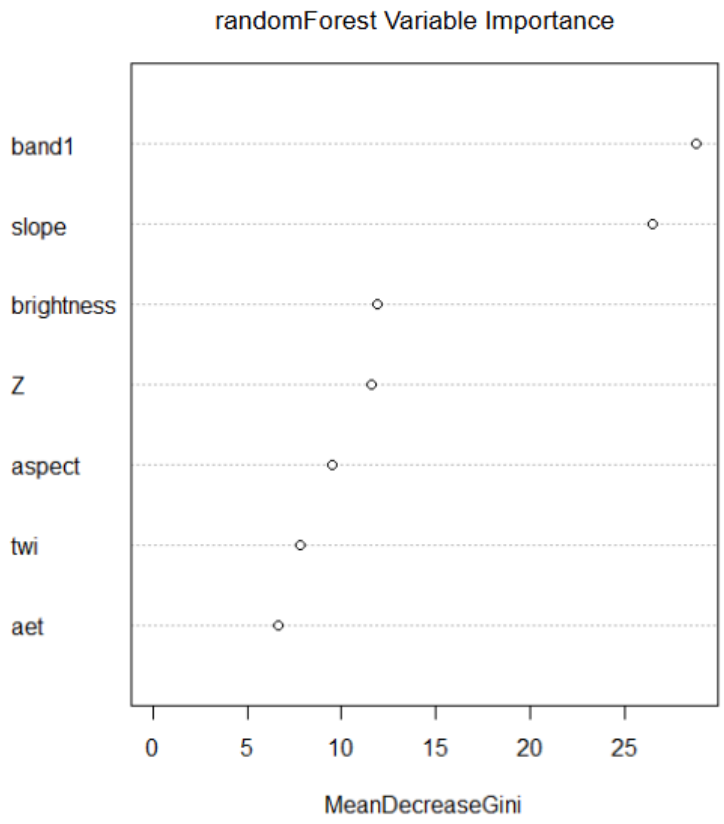

Figure 3. A scatterplot representing the relative importance or weight of each of the seven variables created as background data for pixel-based classification using randomForests.

Error and Error Rate

RandomForest analysis has produced a relatively low classification error rate, with an out-of-bag (OOB) estimate of 14.33%. Further iterations of this methodology would be required to decrease the error rate. Error may also be introduced in the user's choice of class types. For this analysis, only four classes were identified (Figure 1), leading to some land features being combined. For instance, roads and buildings were both considered "developed," and cropland and empty fields were grouped together as "agriculture/grass." Increasing the number of class types may impact the results of a randomForest analysis. In the R script, the most common error users may run into is an inconsistency in raster coordinate reference system (CRS), extent, and resolution. Unification of these values across all rasters is integrated into the script, but manual investigation may be required if an error relating to one of these parameters is discovered. This is an issue unique to the R version of this methodology, as ArcGIS Pro either corrects for these differences automatically or the user is able to correct for them without much issue.

Contour Crop Identification

Remotely identifying croplands that employ contour cropping techniques is straightforward yet time consuming to do manually; however, it may be possible to accomplish this task in an automated fashion. Several attempts have been made at doing just that, primarily using high-pass image filtration. High-pass filters sharpen images, emphasizing the edges of objects. Contour plowing and crop planting results in distinct long, curving strips of crop types. From above, this creates a very obvious striped pattern, as seen in Figure 4. Applying a high-pass filter to an orthoimage allows for the stripes to be isolated as distinct shapes. Automated selection of these stripes is where further research is required. From the filtered image, it is possible to perform image segmentation, resulting in polygons representing land feature shapes. If various high-pass filter kernel sizes and segmentation parameters were explored, it may be possible to produce a polygon feature class that more accurately identifies contour crop stripes. If this were achieved, the resulting polygons could be filtered by their geometries to highlight only the long, thin shapes that correspond with contour crop stripes. Due to the size of the orthomosaic used in this particular project, this methodology was determined to be too time and data-intensive. It may be beneficial to test this methodology on a small section of an orthoimage before applying it to a larger orthomosaic.

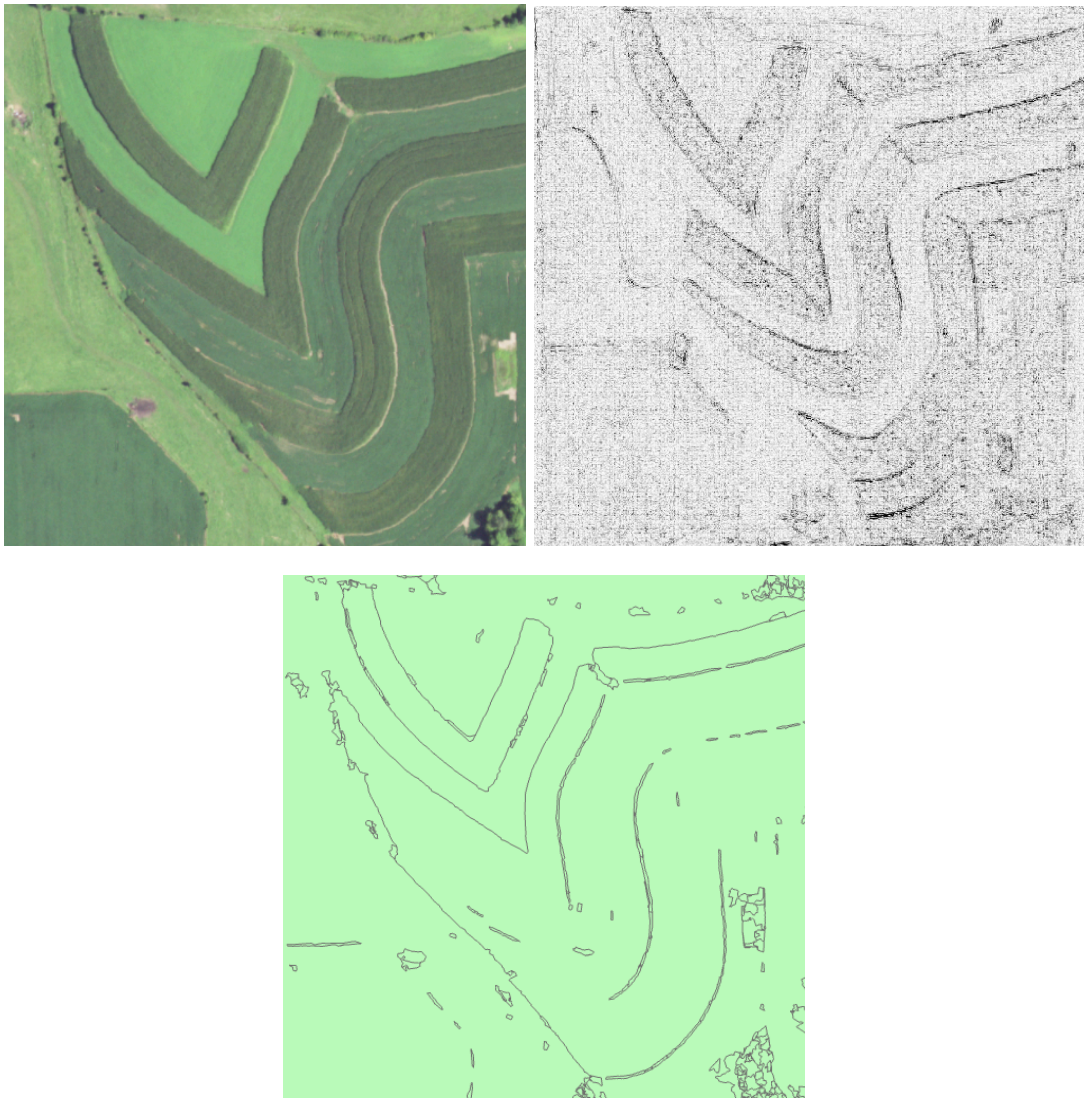

Figure 4. A set of three images showing the same cropland utilizing the contour crop technique. Top left: Unaltered orthoimage, NAIP 2022. Top right: High-pass filter applied. Bottom middle Image has been segmented and transformed into a polygon feature class.

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Potential Other Sources

Paper Methods

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Coding Help/Methods

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